Project Title: Explainable Multi-Agent AI: Generating Human-Readable Decision Traces in Collaborative Agent Systems: A Simple way to Show How Multiple Agents Work Together

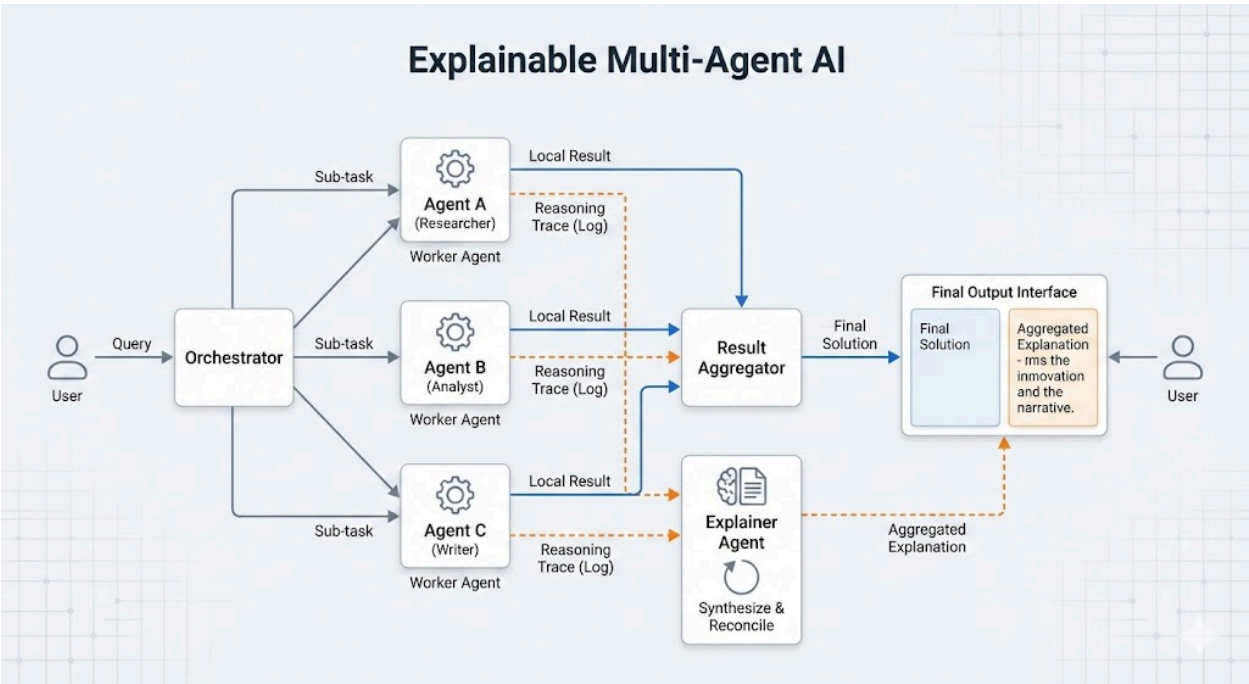
Objectives/Goals:

- Design a simple format so every AI agent in the system can record its own steps.
- Build a specific agent whose only job is to collect everyone's notes and turn them into a single, easy to read explanation.
- Check if these actually help people to trust AI, and make sure the "note-taking" doesn't slow down the system too much.

Expected Outcomes:

- A system with three agents(like a Researcher, an Analyst, and a Writer) that gives you an answer and explains its work
- A way for humans to audit the AI to see exactly where a mistake happened if something goes wrong.

Design Workflow:



- You ask the AI to do something
- A "Boss" agents split the job among three "Worker" agents

- Each Worker does its job and writes down why it made its choices them up into one summary
- The “Explainer Agent” reads all those logs and clean them up into one summary.
- You get the final answer and a clear explanation of how the team got there

Papers:

- **Miller, T. (2019).** *"Explanation in artificial intelligence: Insights from the social sciences."* Artificial Intelligence.
- **Das, S. K., et al. (2021).** *"Explainable Multi-Agent Systems: A Survey.”*
- **Anjomshoae, S., et al. (2019).** *"Explainable Agents and Robots: A Survey of Strategies and Human-Agent Interaction.”*
- **Chan, A., et al. (2023).** *"Governance of Agentic AI Systems."*
- **Bandi, A., et al. (2025).** The Rise of Agentic AI: A Review of Definitions, Frameworks, Architectures, Applications, Evaluation Metrics, and Challenges.